

See the compression damage that metrics miss

Build a VR analysis toolkit for 3D Gaussian splat scenes · Ofinno · CS491 sponsored project

ON A MONITOR
looks fine

IN A HEADSET
holes, floaters,
per-eye mismatch



Same compressed scene. In stereo the damage is obvious.

WHAT YOU BUILD

REVEAL a cut plane you hold in your hand, crop volumes, opacity

RE-RENDER outlines, centers, color the scene by any attribute

QUANTIFY probe one splat, region statistics, histograms

Starting point: SuperSplat, the open-source 3DGS tool we already work with in-house. You do not write a renderer.

HOW SUCCESS IS DEFINED

TIER 1 · it runs in VR
a real scene, in stereo, at interactive frame rate (Fall PoC)

TIER 2 · the MVP
hand-held cut plane, color by attribute, single-splat probe

TIER 3 · your call
pick from our catalog: A/B compare, histograms, floater triage, measurement

WHAT OFINNO SUPPLIES

- A Quest 3S headset on loan for the project
- Real Gaussian splat scenes, original and compressed
- Orientation to the SuperSplat codebase
- Biweekly meetings with our engineers

Graphics, shaders, spatial data, VR interaction: a portfolio project, not a CRUD app.